



Assessment of BCI Performance for Human-Robot Interaction

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ABSTRACT

Brain-computer interface (BCI) devices can enable individuals suffering from motor impairments to live a more comfortable life. By monitoring brain activity, BCIs translate neural processes to control signals for external devices such as robot arms. One method is by detecting and classifying electroencephalography (EEG) signals related to motor effort or execution while minimizing errors due to artifacts. This paper proposes a comprehensive framework for accurate binary motor execution classification between relaxed and activated curl states of the arm using raw EEG signals, using filter bank common spatial pattern (FB-CSP) feature extraction, and machine learning techniques. We tested our approach on 10 healthy subjects aged 22-33 to compare CSP and FB-CSP feature extraction techniques alongside machine learning (ML) techniques of support vector machine (SVM), logistic regression (LR), linear discriminant analysis (LDA) and Naïve Bayes (NB) to identify the best performer. Two types of tests were conducted, including offline and live testing. Human-Machine Interaction (HRI) was accomplished by mapping decision thresholding of the ML probability score outputs to produce a mirrored pose of a robot arm. For our experiments, SVM achieved the highest validation accuracy with an average of 89.09%. Additionally, two types of tests were conducted, including offline and live testing. Furthermore, our results suggest that providing visual feedback is important for motivating and refining motor intent signals for more consistent BCI control. These initial findings highlight the promise of non-invasive BCIs in HRI applications, with ongoing research aimed at further improving accuracy, functionality, and usability to optimally meet the needs of targeted assistive technologies.

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CCS CONCEPTS

• Human-computer interaction; • Machine learning; • Emerging technologies;

KEYWORDS

Brain-Computer Interaction (BCI), Human-Robot Interaction (HRI), Rehabilitation, Electroencephalography (EEG) classification

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1 INTRODUCTION

Brain-computer interface (BCI) devices consist of technologies connecting the human brain and computer systems or other external devices by decoding neural signals. BCI development is driven by their potential to aid the rehabilitation of those with losses in sensorimotor functions [1, 2], with applications extending to gaming, virtual reality, and human-robot interaction (HRI) [3, 4]. BCI can be integrated into a wireless body area network to facilitate convenient and wireless electroencephalography (EEG)-based control of devices such as prosthetics, wheelchairs, or communications ideas [5]. Due to limitations related to nonstationary signal changes, training methods, and information transfer rates, BCI training can be enhanced by multimodal feedback combining visual, auditory, or haptic stimulation, improving user learning to generate consistent control signals [6]. Different motor movements, including single-handed, double-handed, used to operate a BCI system need to be evaluated to ensure accurate performance and user feedback [7].

A key challenge for BCI research is accurately classifying EEG signals associated with motor intent so our areas of interest include enhancing EEG decoding and prediction accuracy, identifying efficient training methods, artifact removal, preprocessing, and classification techniques [8-10]. Preprocessing methods like bandpass filtering can extract frequencies and reduce noise, while Common spatial pattern filtering (CSP) and filter bank common spatial pattern (FB-CSP) [11] are common feature extraction methods to separate

and represent discriminative signal characteristics for classification. Several machine learning (ML) algorithms [8, 12] such as support vector machine (SVM), logistic regression (LR), linear discriminant analysis (LDA), and Naïve Bayes (NB) have been implemented for EEG classification in BCIs.

In our previous work on HRI in [13–15], we tested various HRI schemes for motor rehabilitation exercises with assistive robots for conditions including Cerebral Palsy and Autism Spectrum Disorders. The interface hardware included Electro-Myography (EMG) and RGB-D visual feedback. The goal of this work is to implement an EEG-based BCI framework for controlling a robot manipulator arm. We perform offline processing of calibration EEG data from 10 healthy subjects performing binary motor execution tasks consisting of a relaxed resting state class and an activated curl state and measured the performance of four popular classifiers (SVM, LR, LDA, NB) and two feature extraction methods (CSP, FB-CSP). This framework is used to integrate a robotic arm to mimic the pose of the subject's arm based on ML probability scores. The subjects started with their arm in a relaxed state, the negative class, and then curled their arm up, the positive class. To control the robot the output probability score of the classifier pipeline is used, resulting in the elbow of the robot actuating to a bent position when the probability of the activated state had reached a specific threshold. Once the score fell below the threshold the robot arm returned to the original position at 180-degrees. The operation of a manipulator arm through the EEG interface offers visual feedback to the user for increased awareness and future uses in assistive technologies.

The Key contributions of our work include: 1) Demonstrating that FB-CSP has better performance than CSP for extraction of robust discriminative features for motor effort; 2) Comparing multiple classifiers and establishing FB-CSP+SVM as the best performer; 3) Integrating the classifier with a custom robot arm for HRI testing and visual feedback; 4) Establishing a complete workflow from filtering to classification that achieves over 89% average accuracy across subjects.

The rest of this paper is structured as follows: Section 2 describes the materials and methods used. Section 3 presents and discusses the results. Section 4 concludes the paper, followed by references.

2 MATERIALS AND METHODS

This section outlines the subjects' information, experimental procedure and setup, and the proposed methods for data processing, features extraction, classification, and robot integration.

2.1 Subject Information

Ten healthy subjects 7 males, and 3 females ranging from age 22–33 participated in this experiment after signing informed consent forms.

2.2 Experimental Procedure and Setup

This experiment utilizes OpenBCI's [16] EEG hardware for data acquisition: The Cyton+Daisy board and an electrode cap with 15 electrodes (FP1-FP2, F3-F4, F7-F8, Fz, C3-C4, Cz, P3-P4, Pz, O1-2) capture motor effort, as shown in Figure 1. The board transmits unfiltered EEG data to the OpenBCI GUI for visualization and streaming via Lab Streaming Layer (LSL) [17]. NeuroPype [18] offers

potential for system identification and real-time data processing, including filtering, feature extraction, and machine learning.

The experiment procedure consisted of signal verification, calibration, training, and testing control of an arm robot. After subjects are briefed on the protocol, subjects were fitted with the EEG cap. Signal acquisition was verified through the OpenBCI GUI [16] by testing signal strength and packet loss to ensure the correct placement of each of the electrodes and data integrity before proceeding. Since the EEG signal is transmitted wirelessly, precautions were taken to minimize potential interference by removing nonessential electronics from the experiment area. The experiment then proceeded to calibration data collection. This phase collected subject-specific EEG data during cued motor tasks for developing the classification model. Subjects were seated in front of a set of monitors and asked to familiarize themselves with the muscle movements. Subjects started with their arms resting on their side, acting as the negative class "relaxed." They then physically curled their arm up to act as the positive class, "activated." On-screen text cues prompt the users to perform the relaxed or activated state, streaming cue markers and time stamps to be fused with the EEG data stream by the Lab Recorder program from NeuroPype. This process used 2 rounds of collecting 120 randomly ordered cues of each class. These rounds served as a division of training and testing data for each subject.

Finally, HMI is tested by having the subjects control a robotic arm in live based on the classification of their binary motor execution states. The system's classifier pipeline's output is connected to a remote robot arm. The subject is asked to perform the tasks again to control the arm. When the system output predicts the user is in a relaxed state, the robot actuates to a starting position. When the system output predicts the user is in an activated state the robotic arm rotates to a 90-degree angle. These positions are chosen to mirror the user's pose as a form of visual feedback for control.

2.3 Proposed system

The proposed system is illustrated in the schematic diagram in Figure 2, consisting of signal acquisition, signal processing, classification, and robot integration. The signal processing and classification are implemented in NeuroPype as shown in Figure 3.

2.3.1 Classifier Pipeline. Once the data is recorded, the acquired signals are streamed to NeuroPype using LSL protocol as shown in Figure 3. The marker cue labels are converted to numerical values for use in ML algorithms. The Finite Impulse Response (FIR) Filter is used as a bandpass filter along the time axis, removing unwanted signal components outside 8–30Hz.

The data is then sent into the CSP or FB-CSP node. CSP is an effective feature extraction technique for EEG signals; however, its performance depends heavily on the selection of the optimal frequency band. To address this limitation, FB-CSP applies a filter bank (FB) to decompose the signal into multiple bands before applying CSP to each frequency band individually. The pipeline is tested using SVM, LR, LDA, and NB machine learning techniques to output the probability scores of what class the input is.

2.3.2 Robot Integration. A six-degree-of-freedom (DOF) robotic arm [19] constructed using Dynamixel MX-106 motors is used to

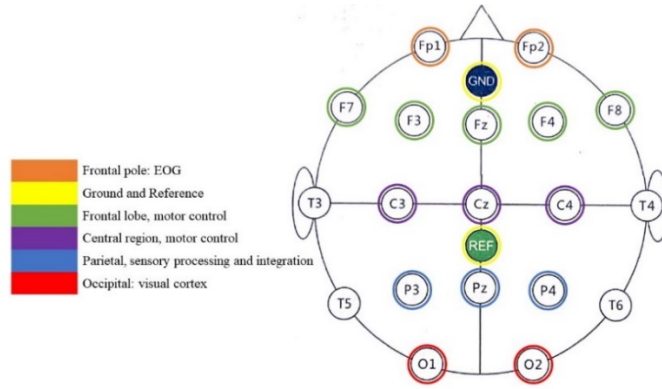


Figure 1: OpenBCI EEG cap and 15 electrode placement montage for our experiments.

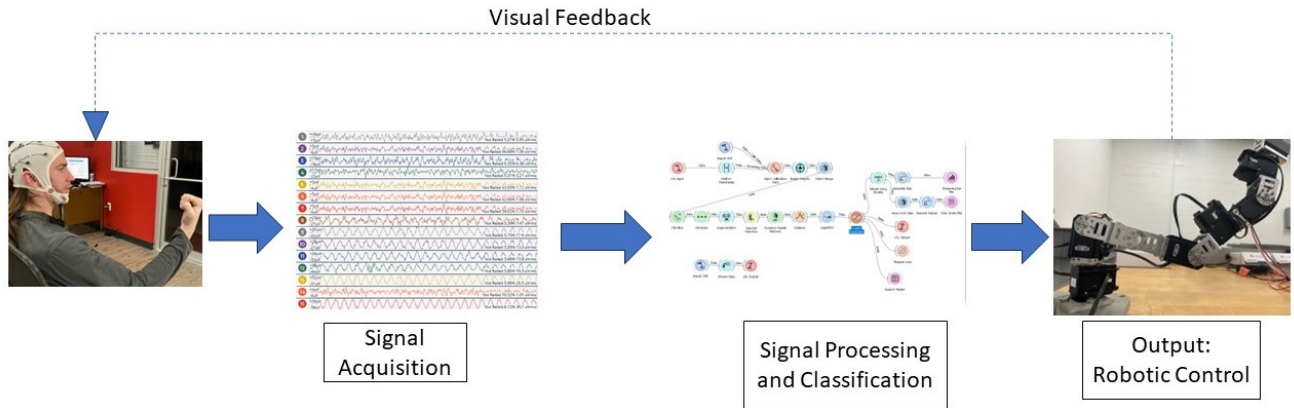


Figure 2: Schematic diagram of our experimental system and data flow, including BCI hardware, data acquisition, processing, classification, and control of a robotic arm.

Table 1: D-H Parameters of our Dynamixel prototype robotic arm.

Joint #	a(mm)	α	d(mm)	Theta
1	0	-90	60.33	Q1
2	161.93	0	0	Q2
3	88.9	-90	0	Q3
4	0	90	60.33	Q4
5	88.9	-90	0	Q5
6	0	0	60.33	Q6

test the ability of the binary classifier to control a robot in a live HMI setting. This robot is intended to be used as a prototype prosthetic arm, featuring a rotating shoulder, hinged elbow, and rotating wrist as shown in Figure 4. The Denavit-Hartenberg (DH) kinematic parameters of the arm are featured in Table 1. The pipeline's outline is transmitted to a robot controller developed using Dynamixel software development kit (SDK) through open sound control (OSC) output protocol.

2.3.3 Control Mapping. User motor executions are mapped to mirroring poses of the robot using the decision thresholding rule of the pipeline's output probabilities. The robot initially assumes the resting state at 180 degrees. Once the probability score for an active state reaches above the threshold of 80% the controller will rotate motor 3, which acts as an elbow, 90 degrees to assume a bent, activated state. If the output probability score decreases below the threshold the controller returns motor 3 to its original position at 180 degrees to reflect a resting state.

Figure 5 shows the block diagram of the entire system as it consists of user signal acquisition from the EEG cap and Cyton+Daisy board interfaced with PC1 containing the OpenBCI GUI and NeuroPy pipeline, connecting over Wi-Fi to a PC2 operating a PID controller. Through USB this controller connects to the robotic arm, where the subject uses it for visual feedback.

3 RESULTS AND DISCUSSION

This paper proposes a comprehensive framework for accurate binary motor execution classification using raw EEG signals. The framework was tested on 10 healthy subjects aged 22-33 to compare the performance of CSP and FB-CSP feature extraction techniques

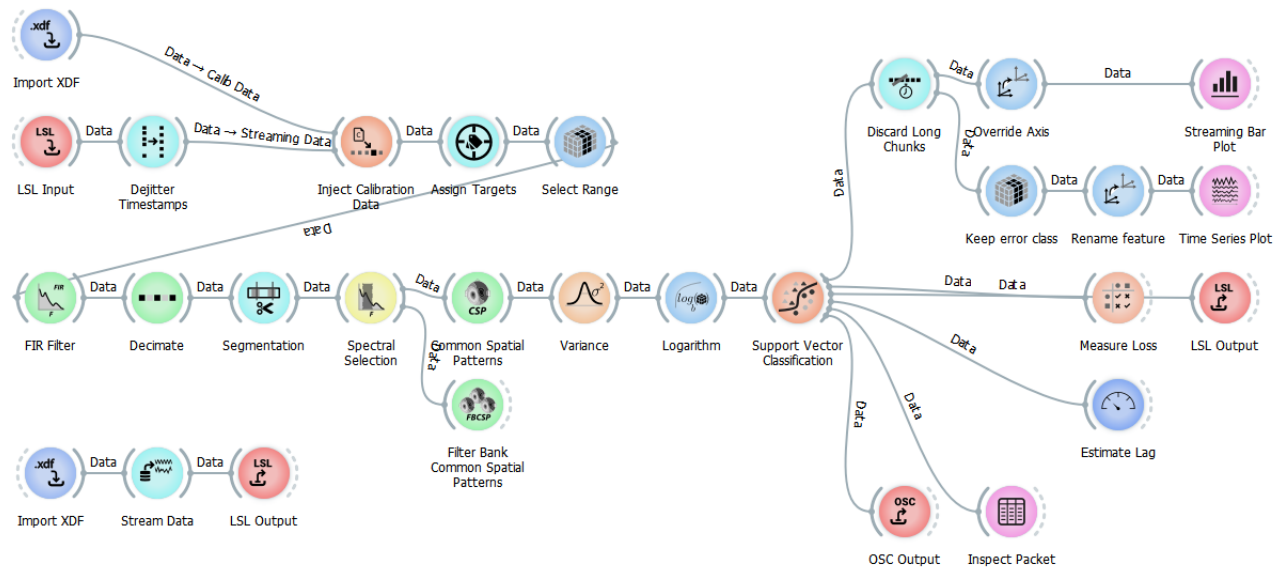


Figure 3: Data pipeline of the proposed system for acquisition, pre-processing, and classification implemented in NeuroPyPe.

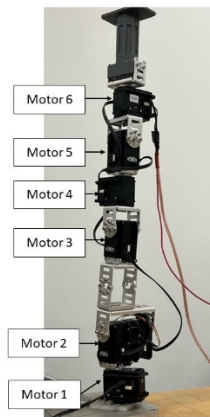


Figure 4: Six DOF robotic arm.

along with the four ML techniques SVM, LR, LDA, and NB. The validation accuracy and AUC of the ML methods were accessed and reported in Tables 2 and 3 for CSP and FB-CSP, respectively. The results demonstrated that FB-CSP outperforms CSP for all of the classifier techniques indicating that that FB-CSP can extract more efficient discriminative spatial patterns compared to CSP using a single band. Overall, SVM+FB-CSP feature extraction produced the top classification performance, achieving $89.09 \pm 7.21\%$ accuracy, showcasing SVM's ability to effectively separate the non-linear separable multiband FB-CSP.

To further test the framework's ability after training to classify new, unseen data, the best-performing feature extract (FB-CSP) with different ML classifiers was tested on 50 trials of data. The test accuracy and AUC for subject A03 are reported in Table 4, showing an accuracy of 86–89% across the classifiers. Figure 6 visualizes

the confusion matrices of the four classifiers for this subject. SVM results in the highest accuracy with 89.8% and AUC of 0.897.

Statistical significance was assessed through the calculation of p-values. The null hypothesis of the p-values in this case would be that there is no difference in performance between feature extraction techniques or the classification methods. One-way ANOVA tests revealed significant differences in performance on the accuracy values and a similar test on the AUC values for each subject. Comparing the classification methods using CSP features, as shown in Table 2, with p-values of 0.027 for accuracy and 0.018 for AUC across the classifiers. Similar ANOVA analyses on FB-CSP features also demonstrated statistically significant disparities in the classifier's performance. To test the statistical significance of the difference between CSP and FB-CSP, a paired t-test was performed on the accuracy and AUC yielded p-values of 0.048 and 0.034 respectively, which is less than the typical significance level of 0.05. This confirmed the superior discriminative ability of FB-CSP over CSP as shown in Tables 3 and 4.

In addition to the offline testing, the framework was integrated with a robot arm to test HRI and provide live visual feedback of the user's arm movement and brain activity. Subjects performed the designated arm movements to activate corresponding robot arm movements. These movements alternate between a relaxed state and an activated curling motion. When the probability score of the activated state class reached 80% threshold, the robotic arm mimicked the pose of the subject, actuating from a pose of 180-degrees upward-facing (relaxed) pose to a predefined 90-degree angle (curled) pose. The live test accuracy was evaluated by running 10 test movements per subject, resulting in an 80% accurate classification. The remaining 20% trials resulted in misclassification due to transient response between the transitions from activated to the relaxed state.

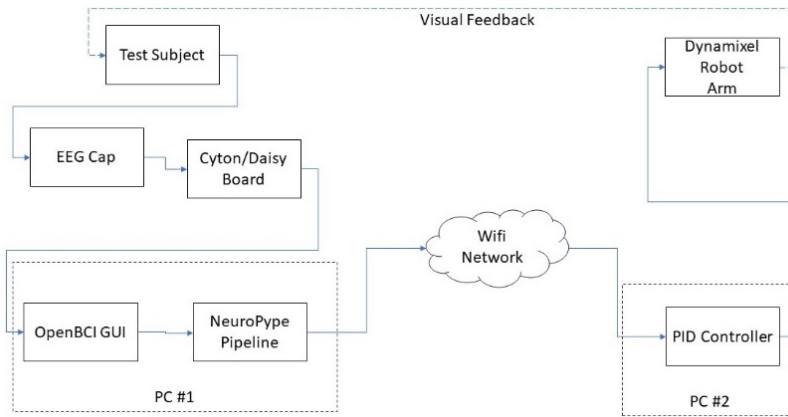


Figure 5: Block Diagram of the complete system featuring two computers connected via WiFi network, and visual feedback to the user.

Table 2: Results of different classifier performances using Common Spatial Pattern (CSP).

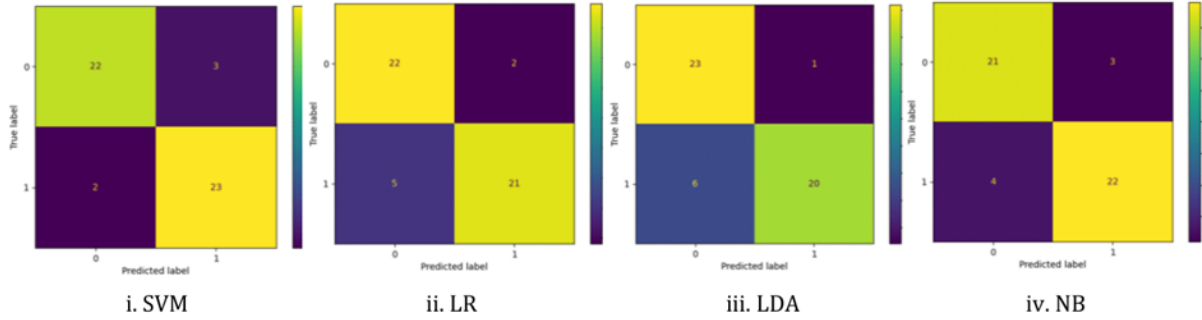
Subject	SVM		LR		LDA		NB	
	Accuracy	AUC	Accuracy	AUC	Accuracy	AUC	Accuracy	AUC
A01	96.7	0.995	98.3	0.93	97.5	0.997	96.7	0.998
A02	88.3	0.942	86.7	0.946	87.5	0.95	90.8	0.959
A03	92.5	0.982	94.2	0.977	95	0.984	94.2	0.967
A04	90.8	0.957	70.8	0.78	69.2	0.751	77.5	0.794
A05	81.7	0.892	82.5	0.886	84.2	0.914	82.5	0.906
A06	99.2	0.999	100	1	99.2	1	99.2	1
A07	77.5	0.842	74.2	0.834	78.3	0.843	74.2	0.835
A08	67.5	0.731	72.5	0.778	69.2	0.781	70.8	0.85
A09	81.7	0.888	83.3	0.924	81.7	0.921	80.8	0.906
A10	85	0.882	83.3	0.925	84.2	0.926	83.3	0.911
Mean $\pm$ Std	86.09 $\pm$ 9.50	0.91 $\pm$ 0.08	84.58 $\pm$ 10.40	0.90 $\pm$ 0.08	84.60 $\pm$ 10.64	0.91 $\pm$ 0.09	85.00 $\pm$ 9.76	0.91 $\pm$ 0.07

Table 3: Results of different classifier performances using FB-CSP.

Subject	SVM		LR		LDA		NB	
	Accuracy	AUC	Accuracy	AUC	Accuracy	AUC	Accuracy	AUC
A01	97.5	0.992	97.5	0.999	97.5	0.992	97.5	0.994
A02	89.3	0.935	89.2	0.939	86.7	0.951	90.8	0.962
A03	95.8	0.991	95	0.982	94.2	0.975	95	0.986
A04	93.3	0.994	77.5	0.851	75.8	0.841	78.3	0.87
A05	83.3	0.913	87.5	0.946	86.7	0.939	88.3	0.945
A06	99.2	1	97.5	1	97.5	0.999	97.5	1
A07	76.7	0.868	77.5	0.855	76.7	0.858	77.5	0.875
A08	84.2	0.929	73.3	0.812	75	0.803	62.5	0.794
A09	85.8	0.906	82.5	0.921	81.7	0.919	83.3	0.926
A10	85.8	0.915	85	0.918	83.3	0.914	80.8	0.912
Mean $\pm$ std	89.09 $\pm$ 7.21	0.94 $\pm$ 0.05	86.25 $\pm$ 8.67	0.92 $\pm$ 0.07	85.51 $\pm$ 8.61	0.92 $\pm$ 0.07	85.15 $\pm$ 10.98	0.93 $\pm$ 0.07

Table 4: Comparison of the average classification test accuracy and AUC for subject A03 using FB-CSP for 50 trials.

Feature Extraction	SVM		LR		LDA		NB	
	Accuracy	AUC	Accuracy	AUC	Accuracy	AUC	Accuracy	AUC
FBCSP	89.8	0.897	86	0.862	85.7	0.859	85.7	0.86

**Figure 6: Single Subject Confusion Matrices for the four classifiers.**

The integration of the robot arm for visual feedback has a crucial role, as it provided motivating and stimulating factors that can improve the user's ability to generate consistent control signals for BCI operation. The live test results demonstrate the feasibility of the proposed approach for intuitive, live control of a robotic arm based on the user's motor intent, paving the way for future applications in assistive technologies and rehabilitation [1, 10].

4 CONCLUSION AND FUTURE WORK

This paper proposed a framework for accurate classification of binary motor effort EEG signals, tailored for the application in HMI systems. Our framework integrates preprocessing, feature extraction, classification techniques, and robot integration. Validation was conducted using data collected from 10 distinct subjects, revealing validation accuracy approaching 90% on average. Notably, the best performance was achieved by the FB-CSP+SVM framework with an average validation accuracy of $89.09 \pm 7.21\%$ and AUC of $0.94 \pm 0.05\%$. Preprocessing includes bandpass filtering from 8-30 Hz using an FIR filter to reduce noise and artifacts. For feature extraction, CSP/FCSP and log-variance are used to extract discriminative spatial patterns and effort-related information. Across our experiment, four classifiers alongside two feature extraction methods were evaluated, with FB-CSP+SVM achieving the highest average accuracy across subjects, thereby providing a slight boost over regular CSP. Importantly our evaluation included two types of tests- offline and live scenario, ensuring the robustness and practical application of our framework. Our future work will explore multi-classification techniques such as deep learning to facilitate graded effort classification and improve performance and usability for motor effort BCIs.

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